

2022-2023学年秋季学期《高等数值分析》期末考试试题

中文版

1. (共10分) 对于线性代数方程组 $Ax = b$, 残量定义为 $r = b - Ax$. 回答下述问题:

- (a) 写出Richardson迭代格式, 分析格式的收敛性.
- (b) 设 M 为可逆矩阵, 写出以 M 为预处理矩阵的Richardson迭代格式.
- (c) 设 A 为对称正定矩阵, 选择预处理矩阵为数乘矩阵, 即 $M = \mu I$ (I 为单位矩阵, μ 为标量). 则当(且仅当) μ 满足何种条件时, 此预处理方法是收敛的?

2. (共10分) 给定系数矩阵为对称正定矩阵的线性代数方程组 $Ax = b$. 请回答下述问题:

- (a) 定义函数

$$\varphi(x) = \frac{1}{2}(Ax, x) - (b, x),$$

其中内积 $(\cdot, \cdot)$ 为标准的欧式内积. 证明 φ 在全空间中存在唯一的临界点; 该临界点为全局最小值点, 且恰好是 $Ax = b$ 的根.

- (b) 设当前近似向量为 x . 写出从 x 出发, 平行于方向 p 做一维极小搜索得到的新近似向量的表达式.
- (c) 证明新近似向量的残量与搜索方向 p 是正交的(关于欧式内积).

3. (共10分) 设 A 是一个实对称矩阵, 则其特征值和特征向量都是实的. 若特征向量 x 满足 $\|x\|_2 = 1$, 则称 x 为归一化特征向量. 请回答下述问题:

- (a) 写出Newton法计算特征值及其对应的归一化特征向量的计算格式.
- (b) 证明当特征值是单特征值时, 上述计算方法是局部二阶收敛的.

4. (共10分) 设实矩阵序列 $\{X_k\}$ 满足

$$X_{k+1} = X_k + X_k(I - AX_k), \quad k = 0, 1, \dots$$

证明如下结论:

- (a) 令 $E_k = I - AX_k$, 则 $E_{k+1} = E_k^2$.
- (b) $\{X_k\}$ 至少局部二阶收敛到 A^{-1} .

5. (共10分) 考虑常微分方程组 $y' = f(t, y)$ 的初值问题. 请回答下述问题:

- (a) 写出一一般步长情况下的显式Euler方法、隐式Euler方法以及梯形方法的计算格式.
- (b) 分析梯形方法的相容性, 写出该方法的主局部截断误差.

6. (共10分) 考虑常微分方程组 $y' = f(t, y)$ 的初值问题. 用如下的等步长Runge-Kutta方法

$$y_{n+1} = y_n + hK,$$

其中 K 满足

$$K = f(t_n + h/2, y_n + hK/2).$$

请回答下述问题:

- (a) 证明该方法恰好是二阶收敛的(需要说明其不是三阶方法).
- (b) 确定该方法的绝对稳定区域.

English Version

1. (10 scores) For the algebraic system $Ax = b$, we define the residual as $r = b - Ax$. Answer the following questions:

- (a) Write out the Richardson iteration scheme and analyze its convergence.
- (b) Let M be an invertible matrix. Write out the Richardson scheme preconditioned by M .
- (c) Suppose A is symmetric and positive-definite. Choose the preconditioner as $M = \mu I$ (I is the identity matrix and μ is a scalar). Determine μ which ensures the convergence of preconditioned Richardson scheme.

2. (10 scores) Given an algebraic system $Ax = b$. Answer the following questions:

- (a) Define

$$\varphi(x) = \frac{1}{2}(Ax, x) - (b, x),$$

where $(\cdot, \cdot)$ indicates the Euclidean inner product. Show that φ admits a unique critical point. In addition, show that this critical point is the global minimizer of φ and solves $Ax = b$.

- (b) Let x be an approximation of exact solution. Write out the updated approximation obtained by the 1D subspace searching on the line which passes through x and orients in the direction p .
 - (c) Show that the residual associated with the updated approximation is orthogonal to the searching direction p with respect to the Euclidean inner product.
3. (10 scores) Let A be a prescribed real symmetric positive-definite matrix. It is known that all eigenvalues are real. An eigenvector x is called normalized if $\|x\|_2 = 1$. Please answer the following question:
- (a) Write out the Newton iteration scheme which simultaneously determines a specific eigenvalue λ and its normalized eigenvector x .
 - (b) Show that when the eigenvalue λ is simple, i.e., with multiplicity of one, the above scheme is locally second-order convergent.
4. (10 scores) Suppose the sequence of real matrix $\{X_k\}$ satisfies

$$X_{k+1} = X_k + X_k(I - AX_k), \quad k = 0, 1, \dots$$

Show the following:

- (a) Set $E_k = I - AX_k$, then it holds that $E_{k+1} = E_k^2$.
 - (b) $\{X_k\}$ locally converges to A^{-1} with second order.
5. (10 scores) Consider the initial value problem of first-order ODE $y' = f(t, y)$. Please answer the following questions:
- (a) Write out the Euler scheme, the backward Euler scheme and the trapezoidal scheme. Here the time-step is not assumed to be constant.

- (b) Analyze the consistency of trapezoidal scheme and write out the dominating term of local truncation error.
6. (10 scores) Consider the initial value problem of first-order ODE $y' = f(t, y)$. We apply the following fixed time-step Runge-Kutta scheme

$$y_{n+1} = y_n + hK,$$

where K solves

$$K = f(t_n + h/2, y_n + hK/2).$$

Please answer the following questions:

- (a) Show that the above scheme is of second-order, but not higher.
- (b) Determine the region of absolute stability.